

Graphite Sensor Test Bench

General Description

This document aims to describe the technical properties, components, and characterization of a graphite sensor using a zero-drift amplifier and analog readings from an Arduino UNO R3. The system uses a digital potentiometer to adjust the sensor gain. The response of the system to gain variations is discussed at the end of the document.

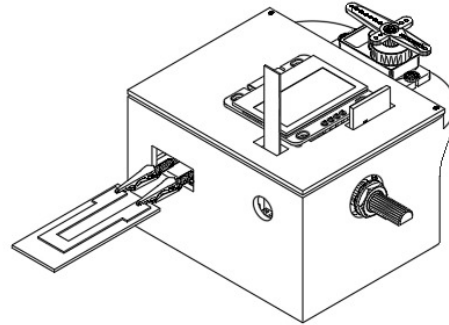


Figure 1: 3D design of the sensor, perspective 1

Applications

Deformation sensors are used in many different fields, allowing them to be applied in various spaces due to the different geometries they can adopt. The present sensor may be useful for:

- Electronic component datasheets
- Technical sales brochures
- Functional specifications

Main Features

- Low-tech design
- Open-source approach
- Practical implementation
- Portable system
- Adjustable response

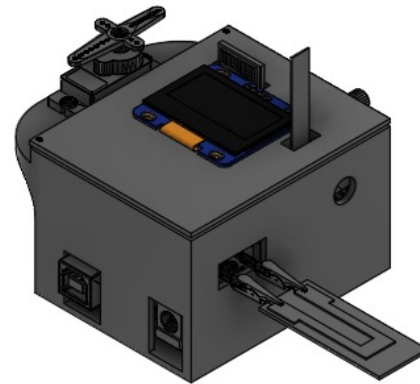


Figure 2: 3D design of the sensor, perspective 2

Active Components

The sensor is composed of the following active components, which are involved in the different functions of the shield [3], [4]. All specifications are given for $-40^{\circ}\text{C} \leq T_A \leq 85^{\circ}\text{C}$, unless otherwise noted.

Table 1: Main Components Used in the Test Bench

Name	Reference	Component Type
Precision Zero-Drift Operational Amplifier	LTC1050 (U2)	Active
Digital Potentiometer	MCP41010 (U3)	Active
Rotary Encoder	KY-040 (ENC1)	Module
Bluetooth Module	HC-05 (U1)	Module
OLED Display	OLED1	Module
Graphite Sensor	GS1	Sensor
Flex Sensor	FL1	Sensor
Servomotor	MS-12016 (M1)	Actuator

The selection of these components is based on their practicality for implementation in an Arduino test bench, as well as their ease of use for the user.

Pinouts

The pinout distribution of the shield is presented in the following table. These choices were made to reduce the distance between the components and their corresponding nodes. At the same time, the manipulable components were placed close to the edges of the shield so that they can be easily accessed by the user when the shield is placed inside the protection box [1].

Arduino Pin	Connections
D2	CLK KY-040
D3	SW KY-040
D4	DT KY-040
D7	TXD HC-05
D8	RXD HC-05
D9	Servomotor Control
D10	CS MCP41010
D11	SI MCP41010
D13	SCK MCP41010
A0	C3, R8
A1	Flex Sensor, R6

Graphite Sensor Dimensions

Ideally, the graphite sensor measures 5 mm in height and 35 mm in width.

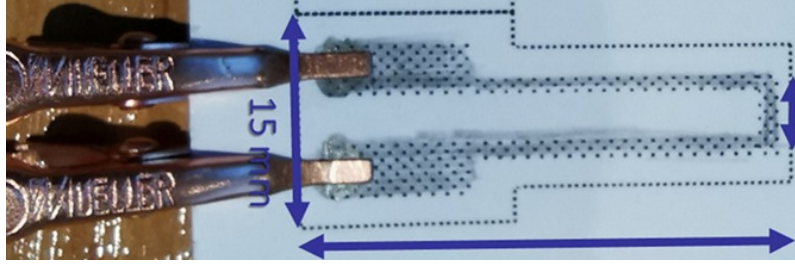


Figure 3: 3D design of the graphite sensor

Different pencil grades contain different amounts of graphite that can be deposited on the substrate. For example, 3B pencils facilitate the detection of resistance variations in the sensor.

$$R_{cap} = \left(1 - \frac{R_4}{R_2}\right) \frac{R_1 V_{cc}}{V_{ADC}} - R_1 - R_3 \quad (1)$$

$$V_{ADC} = \left(1 - \frac{R_4}{R_2}\right) \frac{R_1 V_{cc}}{R_{cap} + R_1 + R_3} \quad (2)$$

$$V_{ADC} = \left(1 - \frac{100k\Omega}{R_{var}}\right) \frac{100k\Omega}{R_{cap} + 100k\Omega} 5V \quad (3)$$

The equation can also be rearranged to express the sensor resistance as a function of the remaining variables:

$$R_{sensor} = \left(1 + \frac{R_4}{R_2}\right) \left(\frac{R_1 V_{cc}}{V_{ADC}}\right) + R_1 + R_3 \quad (4)$$

Signal Filtering

There is one fundamental parameter that can be adjusted in order to reduce the noise in the sensor signal: the value of C_4 . According to Equation (5):

$$f_c = \frac{1}{2\pi R_3 C_4} \quad (5)$$

The value of C_4 determines the effectiveness of the filtering for signals above 50 Hz. This behavior, including the effect of R_2 on the amplifier gain, is illustrated in Figure 4.

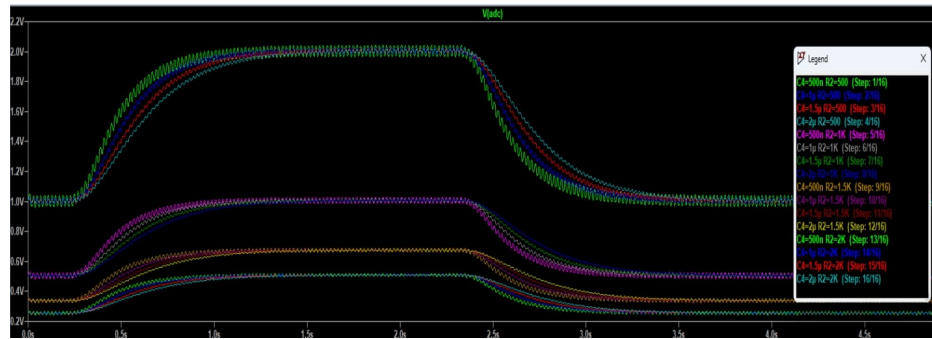


Figure 4: Influence of the filter capacitor and gain adjustment on the sensor response

PCB Design

The distribution of the components on the shield is presented in Figure 5, and a 3D view is illustrated in Figure 6.

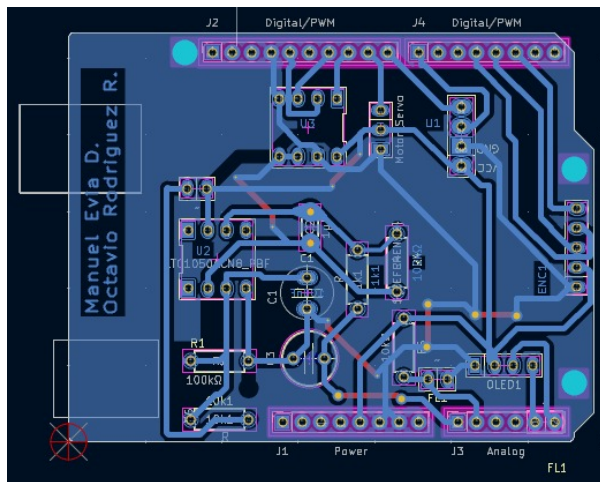


Figure 5: PCB layout of the shield

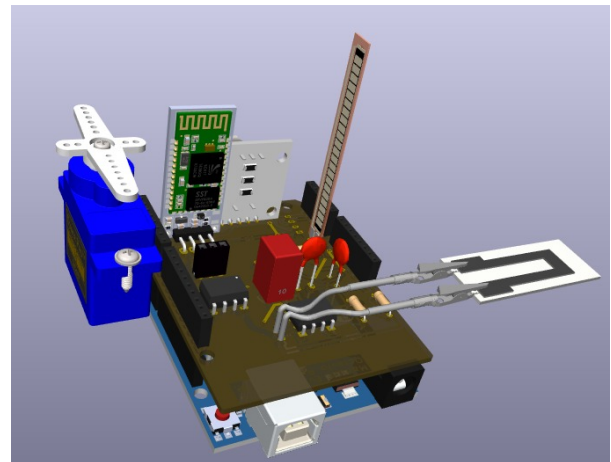


Figure 6: 3D view of the PCB shield

System Response to Graphite Sensor Manipulation

The characterization of the sensor response to mechanical manipulation can be represented using a deformation reference. In this case, a curved solid was used to determine the maximum voltage reading with a multimeter. This value was then compared with the same experiment performed using different R_2 values and different pencil grades, as shown in Figure 7 and Figure 8.

The voltage response increases with the value of R_2 , which is consistent with Equation (3).

System Response to Flex Sensor Manipulation

This system includes a flex sensor that allows the graphite sensor behavior to be compared with that of a commercial sensor that also provides a resistance variation. Figure 9 shows the behavior of this sensor and its linear trend according to the measurements performed with it [2].

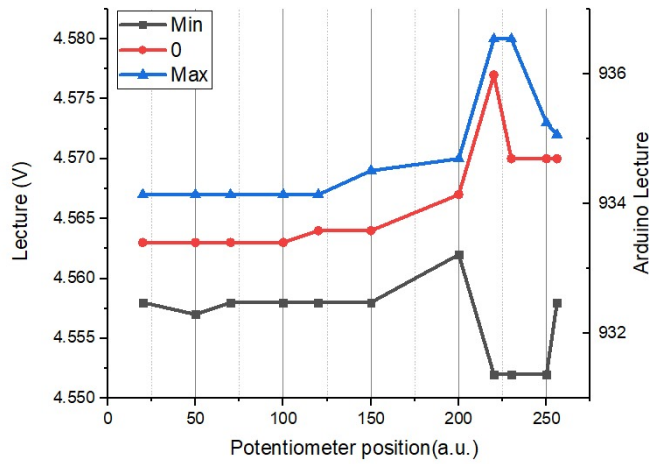


Figure 7: Sensor behavior using a B pencil

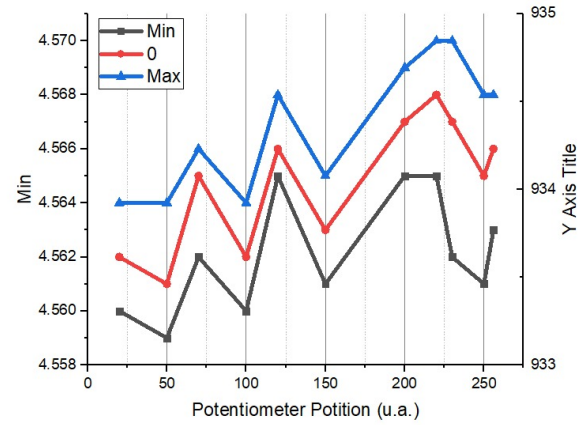


Figure 8: Sensor behavior using a 3B pencil

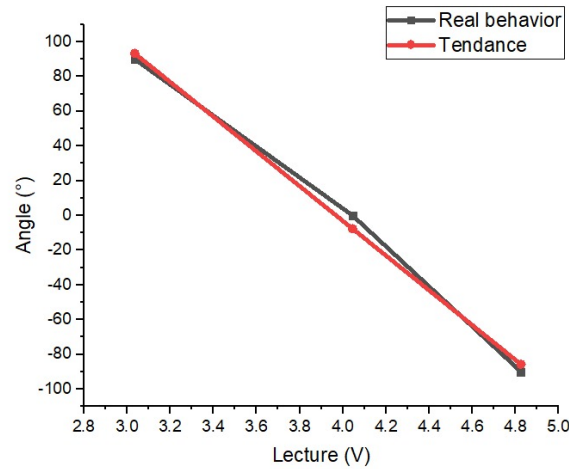


Figure 9: Resistance response of the commercial flex sensor

Limitations

The principal drawback of this sensor is its limited durability and repeatability. Successive measurement cycles lead to the gradual loss or redistribution of graphite particles, resulting in variations of the reference voltage V_{A0} measured by the Arduino, or equivalently in the calculated value of R_{sensor} when the sensor is in its undeformed state. Consequently, the sensor's electrical response becomes less reproducible over time.

Although its lifetime was not quantitatively evaluated, noticeable degradation was observed after only a few measurement cycles, requiring the reapplication of graphite to restore functionality. For applications involving frequent mechanical deformation, a commercial flex sensor may therefore be more suitable due to its longer operational lifetime and improved repeatability.

References

- [1] J. Grisolia, *Microcontrollers & Open Source Hardware*, Département de Génie Physique, INSA Toulouse, Course material, Version 08/04/2024.
- [2] Spectra Symbol, *Flex Sensor FS – Special Edition Length*, Datasheet, Rev. A2, Available at: www.spectrasymbol.com.
- [3] Linear Technology Corporation, *LTC1050 Precision Zero-Drift Operational Amplifier with Internal Capacitors*, Datasheet, Linear Technology Corporation.
- [4] Microchip Technology Inc., *MCP41XXX/42XXX Single/Dual Digital Potentiometer with SPI Interface*, Datasheet, DS11195C, 2003.